# Physics 5C Discussion (Week 10)

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Jinyan has been addicted to playing the game called Factorio with his friends, where they build automated factories. One late-game resource is Uranium: Uranium processing mostly produces the isotope $^{238}_{92}\mathrm{U}$, which is essential in the production of nuclear fuel and "atomic bombs."

**Part (a)** In an atom of $^{238}_{92}\mathrm{U}$, how many electrons, protons, and neutrons does it have?

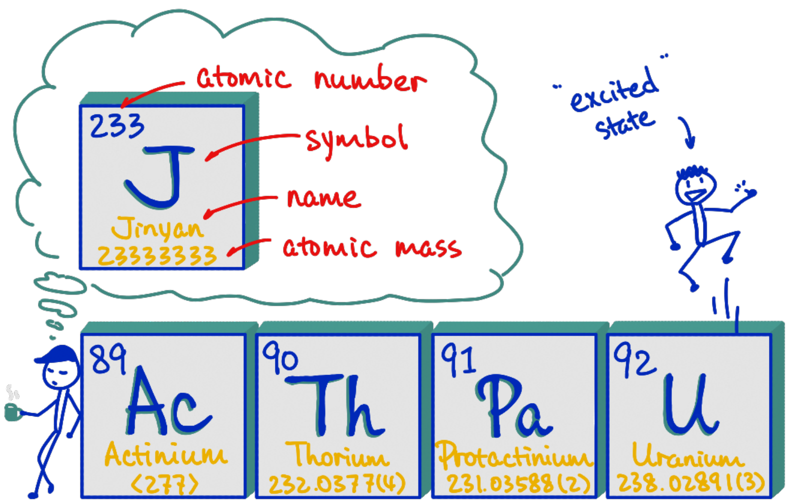
If all of the protons, electrons, and neutrons in the $^{238}_{92}\mathrm{U}$ atom were separated, their total mass is just the sum of the individual masses, but when those particles are bound together to form the $^{238}_{92}\mathrm{U}$ atom, the atom has a different mass. [1]

**Part (b)** Find the total mass of the separated protons and neutrons, from the $^{238}_{92}\mathrm{U}$ atom.

**Part (c)** Find the total mass of the separated electrons, from the $^{238}_{92}\mathrm{U}$ atom.

**Part (d)** Is the mass of the electrons important?

**Part (e)** Calculate the mass defect and the binding energy of $^{238}_{92}\mathrm{U}$.




[1] Here are some constants for your computation:

$$m_p = 1.007276\,\mathrm{u}\,, \qquad m_n = 1.008665\,\mathrm{u}\,, \qquad m_e = 0.000549\,\mathrm{u}\,,$$
$$m_{^{238}_{92}\mathrm{U}} = 238.050788\,\mathrm{u}\,, \qquad h = 6.63\times10^{-34}\,\mathrm{J\cdot s}\,, \qquad c = 3.0\times10^{8}\,\mathrm{m/s}\,.$$

The $^{238}_{92}$U atom is unstable: It will undergo a atomic decay chain, releasing energy in this process (and this is why we say that $^{238}_{92}$U is radioactive), until it ends up at the stable isotope $^{206}_{82}$Pb. The decay chain is a combination of $\alpha$, $\beta$, and $\gamma$ decays.

**Part (f)** Here are some decays in the decay chain. Please help Jinyan to fill in the blanks and complete the decay process. [2]

1. The $^{238}_{92}$U atom undergoes an $\alpha$ decay to $^{234}_{90}$Th. The process is written as

$$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} \ + \alpha\,.$$

2. The $^{234}_{90}$Th atom undergoes a $\beta$-minus decay to ______. This is

$$^{234}_{90}\mathrm{Th} \rightarrow \underline{\qquad\quad} + \underline{\qquad} + \bar{\nu}_e\,.$$

3. The ______ atom undergoes a $\beta$-minus decay to $^{234}_{92}$U. This is

$$\underline{\qquad\quad} \rightarrow {}^{234}_{92}\mathrm{U} + \underline{\qquad} + \bar{\nu}_e\,.$$

4. The $^{234}_{92}$U atom undergoes an $\alpha$ decay to ______. This is

$$^{234}_{92}\mathrm{U} \rightarrow \underline{\qquad\quad} + \underline{\qquad}\,.$$

$$\vdots$$

end. The $^{210}_{84}$Po atom undergoes an ____ decay to the stable $^{206}_{82}$Pb. This is

$$^{210}_{84}\mathrm{Po} \rightarrow {}^{206}_{82}\mathrm{Pb} + \underline{\qquad}\,.$$

**Part (g)** One of the processes in the nuclear decay chain is a $\gamma$ decay,

$$^{214}_{83}\mathrm{Bi}^* \rightarrow {}^{214}_{83}\mathrm{Bi} + \gamma\,.$$

The superscript $*$ is a notation that indicates $^{214}_{83}\mathrm{Bi}^*$ has its nucleus in an excited state, and the nucleus "jump" to the ground state (with the corresponding atom denoted as $^{214}_{83}$Bi) by emitting a $\gamma$-ray. The wavelength of such $\gamma$-ray is measured to be $\lambda = 7.05 \cdot 10^{-31}$ m. Find the energy difference between the $^{214}_{83}\mathrm{Bi}^*$ atom and the $^{214}_{83}$Bi atom in this process.

**Part (h)** Do a quick Google search, and explain why the $\gamma$-ray released in part (g) is highly dangerous to humans. (Hint: What is the frequency of the emitted $\gamma$-ray?) Since the $\gamma$ decay process in part (g) is part of the decay chain of $^{238}_{92}$U, your answer is the main reason why the decay of radioactive substances like $^{238}_{92}$U can "kill" people.

**Part (i)** The half-life of $^{238}_{92}$U is around $4.5 \cdot 10^9$ years. We have 1 gram of pure $^{238}_{92}$U. How long does it take for at least 0.75 grams of it to decay into other substances? [3]

[2] Hint: Check out the cartoon on the front page :)

[3] The decay of a radioactive substance is a random process: A standard nuclear reactor uses 20 tonnes of Uranium a year, which is on the scale of $10^{28}$ Uranium atoms. Therefore, even though the half-life of Uranium seems to be very long, it is still highly likely to see some decay events happening at every single second given the large amount of atoms in the reactor.

## Solution

For this worksheet, the key ideas are the bookkeeping of protons, neutrons, and electrons in an atom, the idea of mass defect, the rules for $\alpha$, $\beta$-minus, and $\gamma$ decays, and the relation between photon energy, wavelength, and frequency. We will use the constants

$$m_p = 1.007276\,\mathrm{u}\,, \qquad m_n = 1.008665\,\mathrm{u}\,, \qquad m_e = 0.000549\,\mathrm{u}\,,$$
$$m_{^{238}_{92}\mathrm{U}} = 238.050788\,\mathrm{u}\,, \qquad h = 6.63 \times 10^{-34}\,\mathrm{J\cdot s}\,, \qquad c = 3.0 \times 10^{8}\,\mathrm{m/s}\,.$$

**Part (a)** The atomic number 92 in the notation $^{238}_{92}\mathrm{U}$ for Uranium tells us that the Uranium atom has 92 protons. Since the atom is charge-neutral, it must also have 92 electrons. The mass number 238 in $^{238}_{92}\mathrm{U}$ is the total number of protons and neutrons, so the number of neutrons is $238 - 92 = 146$. Therefore, an atom of $^{238}_{92}\mathrm{U}$ has

$$92 \text{ protons}, \qquad 92 \text{ electrons}, \qquad \text{and } 146 \text{ neutrons}\,.$$

**Part (b)** The total mass of the separated protons and neutrons is

$$\begin{aligned} m_{p+n} &= 92m_p + 146m_n \\ &= 92(1.007276\,\mathrm{u}) + 146(1.008665\,\mathrm{u}) \\ &= 239.934482\,\mathrm{u} \approx 239.93\,\mathrm{u}\,. \end{aligned}$$

**Part (c)** The total mass of the separated electrons is

$$m_e^{(\mathrm{total})} = 92m_e = 92(0.000549\,\mathrm{u}) \approx 0.05\,\mathrm{u}\,.$$

**Part (d)** For a rough estimate, the electron mass is not very important compared with the proton-and-neutron mass. Indeed,

$$0.05\,\mathrm{u} \ll 239.93\,\mathrm{u}\,.$$

So the electrons contribute only a tiny fraction of the total separated-particle mass. However, if we want the precise mass defect of the whole neutral atom, then we should still include the electrons, because the quoted atomic mass of $^{238}_{92}\mathrm{U}$ already includes the mass of the electrons.

**Part (e)** The mass defect is

$$\Delta m = (\text{mass of separated particles}) - (\text{mass of the bound atom})\,.$$

From parts (b) and (c), the total mass of all separated particles is

$$m_{\mathrm{sep}} = 239.934482\,\mathrm{u} + 0.050469\,\mathrm{u} = 239.984951\,\mathrm{u}\,.$$

Therefore, the mass defect is

$$\Delta m = 239.984951\,\mathrm{u} - 238.050788\,\mathrm{u} = 1.934163\,\mathrm{u} \approx 1.93\,\mathrm{u}\,.$$

Now we turn this mass defect into binding energy by using Einstein's relation

$$E = (\Delta m)c^2\,.$$

The mass $\Delta m$ in this equation should be written in the unit of kg, thus we will need the unit conversion $1\,\mathrm{u} \approx 1.66 \cdot 10^{-27}\,\mathrm{kg}$. Then the binding energy of $^{238}_{92}\mathrm{U}$ is

$$\mathrm{BE} = (\Delta m)c^2 = (1.934163\,\mathrm{u})\,(1.66 \cdot 10^{-27}\,\mathrm{kg/u})\,(3 \cdot 10^{8}\,\mathrm{m/s})^2 = 2.88 \times 10^{-10}\,\mathrm{J}\,.$$

If we want to write it in the unit of MeV, this is $\mathrm{BE} = 1.80 \times 10^{3}\,\mathrm{MeV}$.

**Part (f)** The useful bookkeeping rules are:

($\alpha$). In an $\alpha$ decay, mass number decreases by 4, atomic number decreases by 2.

($\beta$). In a $\beta^-$ decay, mass number stays the same, atomic number increases by 1.

Using those rules, the completed decay processes are listed as follows:

1. Starting with an $\alpha$ decay, ${}^{238}_{92}\mathrm{U} \to {}^{234}_{90}\mathrm{Th} + \alpha$.
2. Followed by a $\beta$-minus decay, ${}^{234}_{90}\mathrm{Th} \to {}^{234}_{91}\mathrm{Pa} + e^- + \bar{\nu}_e$.
3. Followed by a $\beta$-minus decay, ${}^{234}_{91}\mathrm{Pa} \to {}^{234}_{92}\mathrm{U} + e^- + \bar{\nu}_e$.
4. Followed by an $\alpha$ decay, ${}^{234}_{92}\mathrm{U} \to {}^{230}_{90}\mathrm{Th} + \alpha$.

$\vdots$

end. At the end, an $\alpha$ decay, ${}^{210}_{84}\mathrm{Po} \to {}^{206}_{82}\mathrm{Pb} + \alpha$.

Please be careful about the mass number and the atomic number of the element that you put in. They should be calculated from the bookkeeping rules for $\alpha$ and $\beta$ decays. You can find the name of the element from the periodic table by looking up the atomic number. The $\alpha$ particle stands for ${}^4_2\mathrm{He}$, so you can put either $\alpha$ or ${}^4_2\mathrm{He}$ in the equations for $\alpha$ decays.

**Part (g)** Using the wavelength written in the problem, the emitted $\gamma$-photon has energy

$$E = hf = \frac{hc}{\lambda}\,.$$

So we calculate

$$E = \frac{(6.63 \times 10^{-34}\,\mathrm{J \cdot s})(3.0 \times 10^{8}\,\mathrm{m/s})}{7.05 \times 10^{-31}\,\mathrm{m}} = 2.82 \times 10^{5}\,\mathrm{J}\,.$$

The ${}^{214}_{83}\mathrm{Bi}^*$ atom decays to the ${}^{214}_{83}\mathrm{Bi}$ atom by emitting the photon with energy calculated above. Energy has to be conserved, therefore,

$$\Delta E = 2.82 \times 10^{5}\,\mathrm{J}$$

is the energy difference between the two states of the Bismuth atom. If we convert this into electron-Volts,

$$\Delta E = \frac{2.82 \times 10^{5}\,\mathrm{J}}{1.6 \times 10^{-19}\,\mathrm{J/eV}} = 1.76 \times 10^{24}\,\mathrm{eV}\,.$$

**Part (h)** First find the frequency of the emitted $\gamma$-ray:

$$f = \frac{c}{\lambda} = \frac{3.0 \times 10^{8}\,\mathrm{m/s}}{7.05 \times 10^{-31}\,\mathrm{m}} = 4.26 \times 10^{38}\,\mathrm{Hz}\,.$$

This is an extremely high frequency, which means each photon has extremely high energy. An electromagnetic wave (or photon) with such a high frequency is usually referred as a $\gamma$-ray. The high-frequency nature of $\gamma$-rays is what makes it dangerous to humans. These $\gamma$-rays are called "ionizing radiation." In other words, they can knock electrons out of atoms and molecules inside the human body, similar to the role of $\gamma$-rays in the photoelectric effect, where they knock electrons out of a metal by overcoming the binding energy. Therefore, these $\gamma$-rays can break chemical bonds, damage cells, and damage DNA. Also, $\gamma$-rays are very penetrating, so they can travel deep into the body instead of being stopped at the skin. Therefore, exposure to $\gamma$-rays can lead to severe tissue damage, cell death, mutations, and cancer. This is the main reason radioactive substances such as ${}^{238}_{92}\mathrm{U}$ can be deadly.

**Part (i)** A half-life means the amount that remains is "cut in half" after one half-life period of time. We start with $1\,\mathrm{g}$ of ${}^{238}_{92}\mathrm{U}$. After one half-life, the amount remaining is half of $1\,\mathrm{g}$, which is $0.50\,\mathrm{g}$. After two half-lives, the amount remaining is half of $0.50\,\mathrm{g}$ (that is, the count starts from the remaining part after one half-life), which is $0.25\,\mathrm{g}$. That means the amount that has decayed into other types of substances is

$$1.00\,\mathrm{g} - 0.25\,\mathrm{g} = 0.75\,\mathrm{g}$$

after a period of two half-lives. So we need exactly two half-lives. Since one half-life is $4.5 \times 10^9$ years, the required time is

$$\Delta t = 2 \cdot (4.5 \times 10^9\,\text{years}) = 9.0 \times 10^9\,\text{years}\,.$$

It takes $9.0 \times 10^9$ years for at least $0.75\,\mathrm{g}$ of the original Uranium to decay.


Each of them has a story about Physics that you should know :)